

LASITHA DILSHAN THILAKARATHNA

AI Engineer | Generative AI | RAG | LLM Applications | Agentic AI | Full-Stack Development

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PROFESSIONAL SUMMARY

AI Engineer and Full-Stack Software Engineer with 5 years of experience delivering enterprise web applications, Generative AI solutions, Retrieval-Augmented Generation (RAG) systems, agentic AI workflows and cloud-integrated software platforms. Skilled in Python, FastAPI, Angular, React, LangChain, Hugging Face, vector databases, prompt engineering, REST API development and AI-powered automation. Experienced in building scalable dashboards, document intelligence platforms, compliance review assistants, QA automation tools and semantic search applications for enterprise clients.

CORE TECHNICAL SKILLS

Generative AI and LLM Engineering: OpenAI, Hugging Face, Gemini, AWS Bedrock, LangChain, LLM application development, prompt engineering, RAG, Agentic RAG, multi-agent workflows, semantic search, embeddings, document intelligence, conversational AI

Backend and API Development: Python, FastAPI, Flask, Java Spring Boot, NestJS, REST APIs, microservices, streaming responses, file upload workflows, validation workflows, JSON schema extraction

Frontend Engineering: Angular, React, TypeScript, JavaScript, HTML5, CSS3, SCSS, Angular Material, Bootstrap, Material UI, responsive dashboards, reusable UI components, role-based interfaces

Databases and Vector Stores: MySQL, PostgreSQL, Firebase, SQLite, FAISS, ChromaDB, Pinecone, pgvector, persistent data stores

Cloud, DevOps and Tools: AWS, Google Cloud Platform (GCP), Docker, Git, GitHub, GitLab, Agile delivery, CI/CD collaboration

Mobile Development: Android, Java, Firebase Authentication, Firebase Realtime Database, Cloud Firestore, push notifications

PROFESSIONAL EXPERIENCE

Associate Software Engineer | Virtusa, Colombo, Sri Lanka | 2022 - Present

- Develop enterprise full-stack and Generative AI applications using Angular, React, Python, FastAPI, LangChain, Hugging Face, vector databases and cloud services.
- Build AI-powered dashboards, digital asset libraries, document libraries, semantic search interfaces, workflow monitoring modules and compliance-focused user experiences.
- Design and maintain FastAPI services for conversational AI, agentic workflows, retrieval pipelines, content analysis, validation workflows, file processing and real-time data handling.
- Implement RAG solutions using embedding models, FAISS, ChromaDB, Pinecone, prompt engineering and persistent storage to generate context-aware responses and improve knowledge discovery.
- Collaborate with product, QA, architecture and client teams to deliver GenAI, compliance review, test automation and policy analysis solutions in enterprise environments.

Intern Engineer | Surreytech Consulting Sri Lanka | 2021 - 2022

- Supported full-stack development of a doctor appointment management system using Angular frontend technologies and Java Spring Boot backend APIs.
- Implemented reusable UI components, form validations, API integrations and database-driven features for healthcare scheduling workflows.

AI AND SOFTWARE PROJECTS

VirtusaX - Enterprise Functions AI | 2026 - Present | Virtusa Client Project

Technologies: Angular, Python, FastAPI, RAG, embeddings, FAISS, prompt engineering, semantic retrieval, persistent storage

- Developed an enterprise AI platform enabling intelligent automation, knowledge discovery and business process optimization across multiple organizational functions.
- Enhanced Angular dashboards, digital asset libraries, AI-powered search, workflow monitoring and document intelligence capabilities.
- Built and maintained FastAPI microservices for conversational AI, agentic workflows, semantic retrieval, content analysis and real-time data processing.
- Implemented RAG pipelines using embedding models, FAISS vector databases and prompt engineering to generate accurate, context-aware enterprise insights.

Investment AI-MLR Process | 2025 - 2026 | Virtusa Client Project

Technologies: Angular, Python, FastAPI, RAG, FAISS, embeddings, OCR, transcript extraction, streaming APIs

- Developed a full-stack AI-powered MLR assistant platform for content review, compliance validation and knowledge retrieval.
- Built Angular modules for dashboards, document libraries, video disclaimer analysis, OCR processing, transcript extraction, semantic search and progress tracking.
- Developed FastAPI services for conversational AI, claim and disclaimer management, validation workflows, file uploads, streaming responses and agent-based processing.
- Used RAG, embedding models, FAISS vector stores and prompt engineering to improve compliance efficiency and accelerate regulatory review workflows.

Travel Mate Agentic AI Application | 2025 | Virtusa Agentic AI Hackathon

Technologies: Angular, OpenAI LLMs, Google ADK, multi-agent architecture, Agentic AI

- Led and coordinated a team to develop an Agentic AI-powered travel assistant for itinerary generation, travel recommendations and automated planning workflows.
- Integrated Angular frontend experiences with OpenAI LLMs, Google ADK and multi-agent architectures.

- Secured 1st Place in the Virtusa Agentic AI Hackathon for innovation and technical excellence.

PCG QA Test Automation Suite | 2025 | Virtusa Client Project

Technologies: Generative AI, RAG, LLMs, accessibility testing, WCAG, test artifact generation

- Developed an AI-powered QA automation suite that transforms business requirements and user stories into test cases, test scenarios, acceptance criteria and automation-ready scripts.
- Implemented RAG and LLM workflows for requirement analysis, test artifact generation, validation and reporting.
- Built AI-driven accessibility testing capabilities aligned with WCAG standards to identify usability and compliance issues across web applications.

PCG Policy Document Analyzer | 2025 | Virtusa Client Project

Technologies: Angular, Python, FastAPI, LangChain, Gemini LLMs, Hugging Face embeddings, RAG, semantic search

- Developed an AI-powered policy document analyzer that allows users to upload enterprise policy documents and ask contextual questions through a conversational interface.
- Built RAG architecture using Python, FastAPI, LangChain, Gemini LLMs and Hugging Face embedding models for accurate response generation.
- Added response accuracy analysis, confidence scoring, source citation tracking and semantic search to improve policy accessibility and reduce manual review time.

AI-Powered CV Generator | 2024 | Virtusa Client Project

Technologies: React, FastAPI, LangChain, OpenAI LLMs, RAG, JSON schema extraction, prompt templates

- Built an AI-powered resume transformation platform that converts candidate CVs into standardized Virtusa-branded templates.
- Designed RAG and prompt templates to extract, validate and structure candidate information into predefined JSON schemas.
- Developed ReactJS mapping logic to populate prebuilt CV templates, improving formatting consistency and reducing manual processing effort.

SDLC Automation | 2024 | SLIIT MSc Research Project

Technologies: Python, OpenAI LLMs, LangChain, Hugging Face models, RAG, prompt engineering, workflow orchestration

- Designed and developed an AI-powered SDLC automation platform that transforms requirement documents into structured software artifacts.
- Generated user stories, acceptance criteria, test cases, test scenarios and automation test scripts using Generative AI workflows.
- Improved traceability across SDLC phases and accelerated QA preparation through standardized requirement analysis.

BRD to User Story Generator | 2024 | Virtusa GenAI Hackathon

Technologies: RAG, OpenAI LLMs, Hugging Face embeddings, document extraction, semantic search, prompt engineering

- Developed a GenAI solution that converts Business Requirement Documents into structured user stories and acceptance criteria.
- Implemented document extraction, semantic retrieval and prompt engineering to improve Agile planning efficiency and reduce manual analysis effort.

PCG CAMS Application | 2025 | Virtusa Client Project

Technologies: Angular, TypeScript, .NET APIs, role-based UI, data visualization

- Contributed to frontend development of a compliance and workflow management application using Angular.
- Integrated Angular components with .NET backend APIs, implemented form validations, data visualization and role-based UI functionality.
- Improved usability, performance and maintainability through reusable components and cross-functional collaboration.

After Hour Transport and Food Management System | 2023 | Virtusa Client Project

Technologies: Microsoft Power Apps, SharePoint, React, TypeScript

- Developed an employee self-service solution to manage after-hours food and transportation requests.
- Automated request submission, approvals, tracking and reporting to improve operational efficiency and employee experience.

Protective OpenText Program - Print Pro | 2022 | Virtusa Client Project

Technologies: Angular, TypeScript, Bootstrap, SCSS, CSS, Angular Material

- Contributed to a web-based insurance portal by implementing responsive interfaces, reactive forms, reusable components and API integrations.
- Enhanced user experience, data accuracy, application performance and maintainability across insurance-related workflows.

Doctor Appointment Management System | 2021 | Surreytech Consulting Sri Lanka

Technologies: Angular, TypeScript, Bootstrap, SCSS, CSS, Angular Material, Spring Boot APIs

- Developed a healthcare appointment management platform to streamline doctor scheduling and patient booking processes.
- Integrated frontend features with Spring Boot backend APIs for appointment creation, doctor availability, patient registration and notifications.

ADDITIONAL ACADEMIC AND EARLY PROJECTS

SendGrid Email Spring Boot API Project | 2021 | Spring Boot, SendGrid API, REST APIs, HTML email templates

- Developed a reusable Spring Boot email service for transactional and notification emails using configurable workflows and SendGrid integration.

Social Network Android Application | 2020 | Java, SQLite, Firebase Authentication, Firebase Realtime Database, Cloud Firestore

- Designed and developed an Android social networking application with user profiles, posts, media uploads, friend connections, likes, comments and real-time interactions.

Shoe Shop Web Cart Java Web Application | 2019 | Java web technologies, database integration, session handling

- Developed an e-commerce web application with product browsing, cart management, order processing, authentication and inventory workflows.

Phone Shop Sell_Fonic Java Desktop Application | 2018 | Java desktop development, database connectivity, CRUD operations

- Created a desktop retail management system for inventory, customer records, sales processing, billing and stock tracking.

EDUCATION

MSc in Information Technology | SLIIT, Sri Lanka | 2023 - Present

B.Eng. Software Engineering | IIC University, Cambodia | 2018 - 2021

- Second Upper Division | GPA: 3.56

CERTIFICATIONS

- Career Essentials in Generative AI - Microsoft and LinkedIn, 2023
- Generative AI Hackathon Certificate - Virtusa, 2024
- Custom Generative AI Pathway - Virtusa, 2024
- Certified Generative AI Assisted Engineer - Virtusa, 2024

AWARDS

- 1st Place - Virtusa Agentic AI Hackathon, 2025

PUBLICATION

Optimizing User Experience Through Machine Learning: A Review of Key Tools and Technologies - Thilakarathna L. et al., 2024. Review covering machine learning in UX design, privacy, bias mitigation, PyTorch, TensorFlow and Figma.

REFERENCES

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